Gap Note for the Finite Integer-Distance Cluster Problem

Abstract

This proof-attempt isolates two arithmetic routes toward the finite problem left open in `proofs/claims/integer_distance_preprint.tex`. The first uses a Joukowski map from the unit circle to a noncircular ellipse; the second uses the rectangular hyperbola $xy = 1$. Each route proves the geometric reduction and leaves one explicit arithmetic gap.

1 Provenance and status

This note does not prove the main finite existence problem. Its source-backed input is `proofs/claims/integer_distance_preprint.tex`, section "Finite integral point sets in general position", Assumption "Unbounded integral clusters", also logged as `GAP-IDIST-CLUSTERS-01`. The reductions below are local derived claims: the geometric parts are proved here, while the terminal arithmetic inputs remain open.

source-backed
: The preprint's Assumption "Unbounded integral clusters" asks for, for every n, an n-point planar set with pairwise integer distances, no three collinear points, and no four concyclic points.

derived-claim
: Either Missing Lemma B or Missing Lemma D below implies the source-backed finite cluster assumption.

open gap
: The arithmetic statements `GAP-IDIST-ELLIPSE-B` and `GAP-IDIST-HYPERBOLA-D` are not proved here. Section 8 also records `GAP-IDIST-PARABOLA-E`, a new open parabola route suggested after this note was first drafted.

Write

$$\mathbb{Q}^{\square} = \left\{ q^2 : q \in \mathbb{Q} \right\}$$

for the set of rational squares.

2 The finite target

Definition 1 (Good finite set)**.** A finite set $S \subset \mathbb{R}^2$ is *good* if:

1. for any distinct $P, Q \in S$, the distance $|P - Q|$ is a positive integer;
2. no three points of S are collinear;
3. no four points of S are concyclic.

Lemma 2 (Scaling rational distances)**.** *Let $S \subset \mathbb{R}^2$ be finite. Suppose every distance between two distinct points of S is a positive rational number. Then there is a rational number $M > 0$ such that the dilation $MS = \{MP : P \in S\}$ has all pairwise distances in $\mathbb{Z}_{>0}$. Collinearity and concyclicity relations are unchanged by this dilation.*

Proof. If D, the finite set of pairwise distances in S, is empty, take $M = 1$. Otherwise write each $d \in D$ in lowest terms and let M be a common multiple of all denominators. Then $Md \in \mathbb{Z}_{>0}$ for every $d \in D$. A nonzero homothety maps lines to lines and circles to circles. □

Question 3 (Main finite problem)**.** For every positive integer n, does there exist a good set with exactly n points?

The next two sections reduce this question to arithmetic existence problems.

3 The Joukowski ellipse route

Let

$$\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}.$$

For $\mu \in \mathbb{C}^\times$, define

$$J_\mu(z) = z + \frac{\mu}{z} \qquad (z \in \mathbb{T}).$$

Lemma 4 (Derived claim: Joukowski distance identity)**.** *Let $z_i, z_j \in \mathbb{T}$ be distinct. Then*

$$|J_\mu(z_i) - J_\mu(z_j)| = |z_i - z_j|\,|z_i z_j - \mu|.$$

Proof. We compute

$$J_\mu(z_i) - J_\mu(z_j) = z_i - z_j + \mu\left(\frac{1}{z_i} - \frac{1}{z_j}\right) = (z_i - z_j)\left(1 - \frac{\mu}{z_i z_j}\right).$$

Since $|z_i z_j| = 1$, taking absolute values gives the identity. □

Corollary 5 (Derived claim: rational-distance transfer)**.** *Let $z_1, \ldots, z_n \in \mathbb{T}$ be distinct and let $\mu \in \mathbb{C}^\times$. If*

$$|z_i - z_j| \in \mathbb{Q}_{>0} \quad \text{and} \quad |z_i z_j - \mu| \in \mathbb{Q}_{>0}$$

for all $i \neq j$, then the points $J_\mu(z_1), \ldots, J_\mu(z_n)$ have positive rational pairwise distances.

Proof. The distance identity gives rational positive distances, hence distinct image points. □

3.1 Ellipse normalization

Write

$$\mu = \rho\eta^2, \qquad \rho = |\mu| > 0, \qquad \eta \in \mathbb{T}.$$

If $z = \eta w$, then

$$J_\mu(z) = \eta\left(w + \frac{\rho}{w}\right).$$

Thus it is enough to study

$$E_\rho(w) = w + \frac{\rho}{w} \qquad (w \in \mathbb{T}).$$

Writing $w = e^{\mathrm{i}t}$,

$$E_\rho(w) = (1+\rho)\cos t + \mathrm{i}(1-\rho)\sin t.$$

If $\rho \neq 1$, this parametrizes a nondegenerate noncircular ellipse, up to the harmless rotation by η.

Lemma 6 (Derived claim: no three collinear on the ellipse)**.** *Assume $\rho > 0$ and $\rho \neq 1$. If $w_1, \ldots, w_n \in \mathbb{T}$ are distinct, then no three of the points $E_\rho(w_1), \ldots, E_\rho(w_n)$ are collinear.*

Proof. The image is contained in the real conic

$$\frac{x^2}{(1+\rho)^2} + \frac{y^2}{(1-\rho)^2} = 1,$$

whose two denominators are nonzero. A line intersects this nondegenerate conic in at most two real points. Hence three distinct image points cannot be collinear. □

Lemma 7 (Derived claim: four-cyclicity on the ellipse)**.** *Assume $\rho > 0$ and $\rho \neq 1$. Let $w_1, w_2, w_3, w_4 \in \mathbb{T}$ be distinct. Then the four points*

$$E_\rho(w_1),\ E_\rho(w_2),\ E_\rho(w_3),\ E_\rho(w_4)$$

are concyclic if and only if

$$w_1 w_2 w_3 w_4 = 1.$$

Proof. Write a general circle in complex coordinates as

$$\zeta\overline{\zeta} + A\zeta + \overline{A}\,\overline{\zeta} + B = 0,$$

where $A \in \mathbb{C}$ and $B \in \mathbb{R}$. For $\zeta = E_\rho(w) = w + \rho w^{-1}$ and $|w| = 1$, we have

$$\overline{\zeta} = w^{-1} + \rho w.$$

After substitution and multiplication by w^2, the circle equation becomes

$$\rho w^4 + (A + \rho\overline{A})w^3 + (1 + \rho^2 + B)w^2 + (\rho A + \overline{A})w + \rho = 0.$$

Thus if four distinct parameters lie on one circle, they are the four roots of a quartic whose leading and constant coefficients are both ρ. Their product is therefore 1.

Conversely, assume $w_1 w_2 w_3 w_4 = 1$. Let

$$\prod_{k=1}^{4}(w - w_k) = w^4 - e_1 w^3 + e_2 w^2 - e_3 w + 1.$$

Because all w_k lie on $\mathbb{T}$, conjugating is the same as inverting. Thus $e_3 = \overline{e_1}$. Also $\overline{e_2}$ is the sum of $\frac{1}{w_i w_j}$ over the six unordered pairs. Since $w_1 w_2 w_3 w_4 = 1$, these are exactly the complementary pair-products, so $e_2 \in \mathbb{R}$. Write $e_1 = a + \mathrm{i}b$. The equations

$$A + \rho\overline{A} = -\rho e_1, \qquad \rho A + \overline{A} = -\rho\overline{e_1}$$

are solved by $A = x + \mathrm{i}y$ with

$$x = -\frac{\rho a}{1+\rho}, \qquad y = -\frac{\rho b}{1-\rho},$$

using $\rho > 0$ and $\rho \neq 1$. Set $B = \rho e_2 - 1 - \rho^2$, which is real. The corresponding circle equation pulls back to $\rho \prod_{k=1}^{4}(w - w_k) = 0$. Hence the four image points lie on that circle. □

Corollary 8 (Audit correction: phase in the original variables)**.** *With the normalization $\mu = \rho\eta^2$ and $z = \eta w$, four points $J_\mu(z_a), J_\mu(z_b), J_\mu(z_c), J_\mu(z_d)$ are concyclic if and only if*

$$z_a z_b z_c z_d = \eta^4 = \left(\frac{\mu}{|\mu|}\right)^2.$$

Proof. By Lemma 7, concyclicity is equivalent to $(z_a/\eta)(z_b/\eta)(z_c/\eta)(z_d/\eta) = 1$. Since $\mu/|\mu| = \eta^2$, this is exactly the stated condition. □

Theorem 9 (Derived claim: conditional ellipse construction)**.** *Suppose that for every $n \geq 1$ there are distinct points $z_1, \ldots, z_n \in \mathbb{T}$ and a number $\mu \in \mathbb{C}^\times$ such that:*

1. *$|z_i - z_j| \in \mathbb{Q}_{>0}$ for all $i \neq j$;*
2. *$|z_i z_j - \mu| \in \mathbb{Q}_{>0}$ for all $i \neq j$;*
3. *$|\mu| \neq 1$;*
4. *for all distinct a, b, c, d,*

$$z_a z_b z_c z_d \neq \left(\frac{\mu}{|\mu|}\right)^2.$$

Then good sets exist in every finite cardinality.

Proof. For a fixed n, set

$$S_n = \{J_\mu(z_i) : 1 \leq i \leq n\}.$$

The distance identity gives rational pairwise distances. The ellipse normalization with $\rho = |\mu| \neq 1$ gives no three collinear points. The phase criterion in Corollary 8 gives no four concyclic points. The scaling lemma turns S_n into a good set. □

4 A concrete powers specialization

Let θ satisfy $\sin\theta, \cos\theta \in \mathbb{Q}$, and let

$$u = e^{2i\theta}.$$

Then every power of u has rational real and imaginary parts. More is true: the chord lengths among powers of u are rational.

Lemma 10 (Derived claim: rational chord lengths from a rational angle point)**.** *If $\sin\theta, \cos\theta \in \mathbb{Q}$, then*

$$\left|u^i - u^j\right| = 2\left|\sin((i-j)\theta)\right| \in \mathbb{Q}$$

for all integers i, j, where $u = e^{2i\theta}$.

Proof. The trigonometric identity gives

$$\left|e^{2ii\theta} - e^{2ij\theta}\right| = 2\left|\sin((i-j)\theta)\right|.$$

The number $\sin(k\theta)$ is an integral polynomial expression in $\sin\theta$ and $\cos\theta$, so it is rational. □

Remark 11 (Audit correction: the sample angle). The usable Pythagorean example is

$$\cos\theta = \frac{4}{5}, \qquad \sin\theta = \frac{3}{5}.$$

It gives

$$u = e^{2\mathrm{i}\theta} = \frac{7}{25} + \frac{24}{25}\mathrm{i}.$$

The values 5/4 and 5/3 cannot be sine or cosine values of a real angle.

Lemma 12 (Derived claim: the sample power is not torsion)**.** *For $u = 7/25 + (24/25)\mathrm{i}$, the unit complex number u is not a root of unity.*

Proof. If a root of unity ζ has rational real part, then $\zeta + \zeta^{-1} = 2\,\mathrm{Re}(\zeta)$ is both rational and an algebraic integer, hence an integer. For the present u, this number is 14/25, not an integer. Therefore u is not a root of unity. □

Question 13 (Open gap: corrected Missing Lemma B)**.** For every $n \geq 1$, do there exist an angle θ with $\sin\theta, \cos\theta \in \mathbb{Q}$, a non-torsion point $u = e^{2\mathrm{i}\theta}$, and a rational number $\rho \in \mathbb{Q}_{>0} \setminus \{1\}$ such that

$$|\rho - u^m| \in \mathbb{Q} \qquad \text{for every } m = 3, 4, \ldots, 2n-1?$$

Proposition 14 (Derived claim: Missing Lemma B implies the finite target)**.** *If Question 13 has a positive answer for every n, then good sets exist in every finite cardinality.*

Proof. Fix n, take data from Question 13, and set $z_i = u^i$ for $1 \leq i \leq n$. The non-torsion condition makes the z_i distinct. The preceding chord-length lemma gives $|z_i - z_j| \in \mathbb{Q}_{>0}$. Since $3 \leq i + j \leq 2n - 1$, Question 13 gives $|z_i z_j - \rho| \in \mathbb{Q}_{>0}$. Apply Theorem 9 with $\mu = \rho$.

For the four-cyclicity obstruction, $\mu/|\mu| = 1$. If $z_a z_b z_c z_d = 1$, then $u^{a+b+c+d} = 1$, contradicting non-torsion because $a + b + c + d > 0$. Thus the product obstruction is avoided. □

Question 15 (Stronger fixed-angle target)**.** For the fixed value

$$u = \frac{7}{25} + \frac{24}{25}\mathrm{i},$$

does every n admit $\rho \in \mathbb{Q}_{>0} \setminus \{1\}$ such that

$$\rho^2 + 1 - 2\rho\,\mathrm{Re}(u^m)$$

is a rational square for every $m = 3, 4, \ldots, 2n-1$?

This fixed-angle target is stronger than Question 13. A failure of the fixed-angle target would not by itself refute the moving-u version.

Remark 16 (Audit correction: rational coordinates alone). The condition $u \in \mathbb{T} \cap \mathbb{Q}(i)$ does not by itself imply $|u^i - u^j| \in \mathbb{Q}$ for all i, j. The powers specialization above uses the stronger hypothesis $u = e^{2\mathrm{i}\theta}$ with $\sin\theta, \cos\theta \in \mathbb{Q}$, which directly proves rational chord lengths.

5 The hyperbola route

For $t \in \mathbb{Q}^\times$, define

$$H(t) = \left(t, \frac{1}{t}\right) \in \mathbb{R}^2.$$

Lemma 17 (Derived claim: hyperbola distance formula)**.** *For distinct $s, t \in \mathbb{Q}^\times$,*

$$|H(s) - H(t)| = \frac{|s-t|}{|st|}\sqrt{(st)^2 + 1}.$$

Proof. We have

$$\frac{1}{s} - \frac{1}{t} = -\frac{s-t}{st}.$$

Therefore

$$|H(s) - H(t)|^2 = (s-t)^2 + \left(\frac{1}{s} - \frac{1}{t}\right)^2 = (s-t)^2\left(1 + \frac{1}{s^2t^2}\right),$$

which is the square of the displayed formula. □

Corollary 18 (Derived claim: rational distances on the hyperbola)**.** *Let $t_1, \ldots, t_n \in \mathbb{Q}^\times$ be distinct. If*

$$(t_i t_j)^2 + 1 \in \mathbb{Q}^\square$$

for every $i \neq j$, then the points $H(t_1), \ldots, H(t_n)$ have rational pairwise distances.

Proof. The prefactor $|t_i - t_j| / |t_i t_j|$ is rational and positive. The remaining square-root term is rational by hypothesis. □

Lemma 19 (Derived claim: no three collinear on $xy = 1$)**.** *No line in $\mathbb{R}^2$ contains three distinct points of the hyperbola $xy = 1$.*

Proof. A vertical line intersects $xy = 1$ in at most one point. Any nonvertical line has equation $y = mx + b$. Substitution gives

$$x(mx + b) = 1,$$

a polynomial equation of degree at most two in x. Hence such a line has at most two intersections with the hyperbola. □

Lemma 20 (Derived claim: four-cyclicity on $xy = 1$)**.** *Let $t_a, t_b, t_c, t_d \in \mathbb{Q}^\times$ be distinct. The four points*

$$H(t_a),\ H(t_b),\ H(t_c),\ H(t_d)$$

are concyclic if and only if

$$t_a t_b t_c t_d = 1.$$

Proof. A circle has equation

$$x^2 + y^2 + Ax + By + C = 0$$

for some real A, B, C. Substituting $y = 1/x$ and multiplying by x^2 gives

$$x^4 + Ax^3 + Cx^2 + Bx + 1 = 0.$$

Thus, if four distinct hyperbola parameters lie on one circle, they are the four roots of a monic quartic with constant coefficient 1, so their product is 1.

Conversely, suppose $t_a t_b t_c t_d = 1$. Write

$$\prod_{r\in\{a,b,c,d\}} (x - t_r) = x^4 - e_1 x^3 + e_2 x^2 - e_3 x + 1.$$

Taking $A = -e_1$, $C = e_2$, and $B = -e_3$, the circle equation above pulls back to this quartic. Hence the four corresponding points lie on a circle. □

Corollary 21 (Derived claim: positivity prevents concyclicity)**.** *If $t_1, \ldots, t_n \in \mathbb{Q}$ are all greater than 1, then no four of the points $H(t_1), \ldots, H(t_n)$ are concyclic.*

Proof. The product of any four distinct selected parameters is greater than 1, so Lemma 20 excludes concyclicity. □

Question 22 (Open gap: Missing Lemma D)**.** For every $n \geq 1$, do there exist distinct rationals

$$1 < t_1 < \cdots < t_n$$

such that

$$(t_i t_j)^2 + 1 \in \mathbb{Q}^{\square} \qquad \text{for all } i < j?$$

Theorem 23 (Derived claim: Missing Lemma D implies the finite target)**.** *If Question 22 has a positive answer for every n, then good sets exist in every finite cardinality.*

Proof. For a fixed n, choose $t_1, \ldots, t_n$ as in Question 22 and set

$$S_n = \{H(t_i) : 1 \leq i \leq n\}.$$

The hyperbola rational-distance corollary gives rational pairwise distances. The hyperbola line-intersection lemma gives no three collinear points. Since all $t_i > 1$, the positivity corollary gives no four concyclic points. The scaling lemma gives a good set with n points. □

6 Equivalent forms of Missing Lemma D

Question 22 can be restated in several ways useful for later agents.

Proposition 24 (Derived claim: square-entry form)**.** *Question 22 is equivalent to the following statement: for every $n \geq 1$, there exist distinct rational squares*

$$a_1, \ldots, a_n > 1$$

such that

$$a_i a_j + 1 \in \mathbb{Q}^{\square} \qquad \textit{for all } i < j.$$

Proof. If t_i solve Question 22, set $a_i = t_i^2$. Conversely, if a_i are rational squares greater than 1, choose their positive rational square roots $t_i > 1$. The displayed conditions are then identical. □

Proposition 25 (Derived claim: multiplicative clique form)**.** *Let*

$$P = \left\{x \in \mathbb{Q}_{>0} : x^2 + 1 \in \mathbb{Q}^{\square}\right\}.$$

Question 22 is equivalent to asking for arbitrarily large finite sets $T \subset \mathbb{Q}_{>1}$ such that

$$st \in P \qquad \textit{for all distinct } s, t \in T.$$

Moreover,

$$P = \left\{\frac{1-q^2}{2q} : q \in \mathbb{Q}, 0 < q < 1\right\} = \left\{\frac{2r}{r^2-1} : r \in \mathbb{Q}, r > 1\right\}.$$

Proof. The equivalence with Question 22 follows from taking $T = \{t_1, \ldots, t_n\}$.

For the parametrization, solve $x^2 + 1 = y^2$ with $x, y \in \mathbb{Q}$ and $x > 0$. Then

$$(y - x)(y + x) = 1.$$

Put $q = y - x$. Since $x > 0$ and $y > 0$, one has $0 < q < 1$, and

$$x = \frac{1 - q^2}{2q}.$$

Conversely, this formula gives a positive rational x with $x^2 + 1 = ((1 + q^2)/(2q))^2$. The second parametrization follows by the change of variable $q = (r - 1)/(r + 1)$, where $r > 1$. □

7 Gap register

`GAP-IDIST-CLUSTERS-01`

Source-backed finite obstruction from `proofs/claims/integer_distance_preprint.tex`: construct, for every n, an n-point planar integral cluster with no three collinear points and no four concyclic points.

`GAP-IDIST-ELLIPSE-B`

Open arithmetic input for the ellipse route: prove Question 13, or prove a variant strong enough to satisfy Theorem 9. This is a simultaneous rational-distance problem from one rational real point to finitely many powers on the unit circle.

`GAP-IDIST-HYPERBOLA-D`

Open arithmetic input for the hyperbola route: prove Question 22. Equivalently, construct arbitrarily large rational Diophantine tuples whose entries are rational squares.

`GAP-IDIST-PARABOLA-E`

Open arithmetic input from the parabola route in Section 8: construct arbitrarily large finite sets of positive rationals whose pairwise sums are rational legs of rational right triangles.

`GAP-IDIST-JOUKOWSKI-PHASE-01`

Audit item resolved in Corollary 8: under the normalization $\mu = \rho\eta^2$, the product obstruction in the original z-variables is $(\mu/|\mu|)^2$.

`GAP-IDIST-UNIT-POWERS-01`

Audit item resolved in the powers specialization: rational real and imaginary parts of u alone are not used as the chord-length hypothesis. The note uses the stronger rational-angle condition $u = e^{2i\theta}$ with $\sin\theta, \cos\theta \in \mathbb{Q}$.

8 Next attack target: standalone problem

The most promising first gap to attack is now `GAP-IDIST-PARABOLA-E`. It is not proved here, and the proposed arithmetic-progression/CRT argument for it should not be treated as source-backed. Its advantage is that the geometric verification is elementary and completely separated from the arithmetic input.

Question 26 (Open gap: standalone parabola problem)**.** For every integer $n \geq 1$, do there exist distinct positive rational numbers

$$x_1, \ldots, x_n$$

such that

$$1 + (x_i + x_j)^2 \in \mathbb{Q}^{\square} \qquad \text{for all } i < j?$$

Equivalently, can one find arbitrarily large finite subsets of $\mathbb{Q}_{>0}$ whose pairwise sums are rational numbers s for which $1 + s^2$ is a rational square?

Proposition 27 (Derived claim: the parabola problem implies the finite target)**.** *If Question 26 has a positive answer for every n, then good sets exist in every finite cardinality.*

Proof. Given $x_1, \ldots, x_n$ from Question 26, set

$$P_i = (x_i, x_i^2).$$

For $i \neq j$,

$$|P_i - P_j|^2 = (x_i - x_j)^2(1 + (x_i + x_j)^2),$$

so the pairwise distances are rational. A line meets the parabola $y = x^2$ in at most two points, so no three selected points are collinear. If four selected points lie on a circle

$$x^2 + y^2 + Dx + Ey + F = 0,$$

then substituting $y = x^2$ gives

$$x^4 + (E + 1)x^2 + Dx + F = 0.$$

The four corresponding positive x-coordinates would be the roots of this quartic, whose x^3-coefficient is zero; their sum would be zero, impossible. The scaling lemma then gives a good set. $\square$

A useful but stronger subproblem is to find, for every n, a nonzero rational number d such that

$$1 + m^2 d^2 \in \mathbb{Q}^{\square} \qquad (m = 3, 4, \ldots, 2n - 1).$$

Then $x_i = id$ would solve Question 26. This stronger statement is currently only a candidate attack route; the quoted Chinese-remainder or Kemnitz-style construction is not verified in this note and is false as stated for the sample formula tested during audit.

The hyperbola target `GAP-IDIST-HYPERBOLA-D` remains the next cleanest alternative: find rational numbers $t_i > 1$ with

$$(t_i t_j)^2 + 1 \in \mathbb{Q}^{\square}$$

for all pairs. The ellipse target `GAP-IDIST-ELLIPSE-B` offers more parameters, but its bookkeeping is less convenient for a first standalone attack.